Topic 5: Analysis under the Two-Factor Factorial Design
Chapter 11.7-11.8 of our Textbook
STAT 2509: Introduction to Statistical Modeling II

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The $a \times b$ (Two-Factor) Factorial Design

- We extend the single-factor design to consider **two factors** simultaneously.
- Factor A has a levels: $i = 1, 2, \ldots, a$
- Factor B has b levels: $j = 1, 2, \ldots, b$
- The response variable y is divided into $a \times b$ sub-populations.
- Each (i, j) combination of levels of factors A and B is called a **treatment**.
- The sub population of y corresponding to a treatment is a **treatment population**.
- Its mean of a treatment population is the **treatment mean** or **cell mean** μ_{ij}.
- In the $a \times b$ factorial setup design, we mainly interested in comparing the means of the $a \times b$ sub-populations of y.

Table: Treatment Combinations in an $a \times b$ Factorial Design

A \B	**1**	**2**	$\cdots\cdots$	b
1	Sub-population $(1, 1)$	Sub-population $(1, 2)$	$\cdots\cdots$	Sub-population $(1, b)$
2	Sub-population $(2, 1)$	Sub-population $(2, 2)$	$\cdots\cdots$	Sub-population $(2, b)$
$\vdots$	$\vdots$	$\vdots$	$\cdots\cdots$	$\vdots$
a	Sub-population $(a, 1)$	Sub-population $(a, 2)$	$\cdots\cdots$	Sub-population (a, b)

Two-Factor Factorial Design

A design is called a $a \times b$ **Factorial Design** (or textbfTwo-Factor Factorial Design) if a random sample is selected independently from each of the $a \times b$ treatment population (sub-populations) of y defined by the combinations of levels of factor A (a levels) and levels of factor B (b levels).

Notation

- r = sample size for each treatment (balanced design).
- y_{ijk} = response for the kth observation at level i of A and level j of B.
- $k = 1, \ldots, r; \quad i = 1, \ldots, a; \quad j = 1, \ldots, b.$
- Total sample size: $n = a \times b \times r = abr$.

The Two-Factor Factorial Model

The **full model** of $a \times b$ factorial design with interaction:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where:

- μ: overall mean
- α_i: main effect of factor A at level i (deviation from μ due to A)
- β_j: main effect of factor B at level j (deviation from μ due to B)
- $(\alpha\beta)_{ij}$: interaction effect between A at level i and B at level j
- $\varepsilon_{ijk} \sim N(0, \sigma^2)$: random error, i.i.d.

Note the **treatment mean** (or **cell mean**) for the (i, j)th treatment is given by:

$$\mu_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}$$

Model Assumptions

1. $a \times b$ treatment samples are random samples collected independently from their respective treatment populations (i.e. $a \times b$ factorial design).
2. All treatment populations (Sub-populations) have normal distributions.
3. All treatment populations have equal variance σ^2.
4. $$\sum_{i=1}^{a} \alpha_i = 0, \quad \sum_{j=1}^{b} \beta_j = 0,$$

   $$\sum_{i=1}^{a} (\alpha\beta)_{ij} = 0 \quad \text{for all } j = 1, \cdots, b,$$

   and

   $$\sum_{j=1}^{b} (\alpha\beta)_{ij} = 0 \quad \text{for all } i = 1, \cdots, a.$$

Interpretation and Example

Study plant growth (y) with two factors:

- Factor A: Fertilizer type (Organic, Chemical) $\rightarrow a = 2$
- Factor B: Water frequency (Low, High) $\rightarrow b = 2$
- $r = 5$ plants per treatment combination.

B		Low	High
A	Organic	$y_{11k} = \mu + \alpha_1 + \beta_1 + (\alpha\beta)_{11} + \varepsilon_{11k}$	$y_{12k} = \mu + \alpha_1 + \beta_2 + (\alpha\beta)_{12} + \varepsilon_{12k}$
	Chemical	$y_{21k} = \mu + \alpha_2 + \beta_1 + (\alpha\beta)_{21} + \varepsilon_{21k}$	$y_{22k} = \mu + \alpha_2 + \beta_2 + (\alpha\beta)_{22} + \varepsilon_{22k}$

Where

$$\begin{aligned}
\mu &= \text{ overall mean plant growth} \\
\alpha_1 &= \text{ effect of Organic fertilizer } (\alpha_1 + \alpha_2 = 0) \\
\alpha_2 &= \text{ effect of Chemical fertilizer} \\
\beta_1 &= \text{ effect of Low water frequency } (\beta_1 + \beta_2 = 0) \\
\beta_2 &= \text{ effect of High water frequency} \\
(\alpha\beta)_{ij} &= \text{ interaction effect for combination } i, j \\
\varepsilon_{ijk} &= \text{ random error for plant } k \text{ in treatment } ij \\
k &= 1, 2, \ldots, 5 \text{ (5 plants per treatment combination)}
\end{aligned}$$

For a plant with Chemical fertilizer and Low water:

$$y_{21k} = \mu + \alpha_2 + \beta_1 + (\alpha\beta)_{21} + \varepsilon_{21k}$$

Here α_2 is the effect of Chemical fertilizer, β_1 the effect of Low water, and $(\alpha\beta)_{21}$ the extra effect specific to that combination.

Model Without Interaction

If there is **no interaction** between factors:

$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}$$

Here, the effect of factor A is the same at all levels of B, and vice versa.
The difference between any two cell means depends only on the levels of the factors involved, not their combination. That is:

- The (main) effect of ith level of A (which is the same for all levels of B) is

$$\mu_{i\cdot} = \mu + \alpha_i,$$

- The (main) effect of jth level of B (which is the same for all levels of A) is

$$\mu_{\cdot j} = \mu + \beta_j,$$

Note when there is no interaction and all levels of A has the same effect for all levels of B and vice-versa then

$$\mu_{i\cdot} = \mu_{\cdot j} = \mu$$

Factorial Design vs. Completely Randomized Design (CRD)

$a \times b$ **Factorial Design**	**CRD**
Two (or more) factors studied simultaneously	Only one factor (or multiple factors but treated as one combined factor)
Explicitly models interaction	Cannot model interaction between factors
Efficient use of data	Less efficient for studying multiple factors
Allows separate tests for A, B, and AB	Only one overall treatment effect

Goal of the Analysis

1. Determine if factor A has a significant effect on the response.
2. Determine if factor B has a significant effect on the response.
3. Determine if there is a significant interaction between A and B.
4. Estimate the model parameters and treatment means.
5. Perform multiple comparisons if needed.

Three Sets of Hypotheses to Test

1. **Interaction:**

$$H_0^{AB} : (\alpha\beta)_{ij} = 0 \quad \text{for all } i, j$$
$$H_a^{AB} : \text{ at least one } (\alpha\beta)_{ij} \neq 0.$$

2. **Main effect of A:**

$$H_0^{A} : \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$$
$$H_a^{A} : \text{ at least one } \alpha_i \neq 0.$$

3. **Main effect of B:**

$$H_0^{B} : \beta_1 = \beta_2 = \cdots = \beta_b = 0$$
$$H_a^{B} : \text{ at least one } \beta_j \neq 0.$$

Note: If there is no interaction (i.e., we fail to reject H_0^{AB}), we then proceed to test the main effects of factors A and B by evaluating the hypotheses H_0^A and H_0^B described above. Otherwise, if H_0^{AB} is rejected (i.e., there are interactions), we **SHOULD NOT** test the main effects of factors A and B. Instead, we need to find out how treatment means differ using $\binom{ab}{2} = \frac{ab(ab-1)}{2}$ multiple comparisons by testing the hypothesis $H_0 : \mu_{ij} = \mu_{\ell m}$ versus $H_a : \mu_{ij} \neq \mu_{\ell m}$ for all pairs (i, j) and (ℓm).

Data Layout in Two-Factor Factorial Design Model

	Factor B			
Factor A	**1**	**2**	$\dots$	**b**
1	$\underbrace{y_{111}, y_{112}, \dots, y_{11r}}_{total = y_{11\cdot}}$	$\underbrace{y_{121}, y_{122}, \dots, y_{12r}}_{total = y_{12\cdot}}$	$\dots$	$\underbrace{y_{1b1}, y_{1b2}, \dots, y_{1br}}_{total = y_{1b\cdot}}$
2	$\underbrace{y_{211}, y_{212}, \dots, y_{21r}}_{total = y_{21\cdot}}$	$\underbrace{y_{221}, y_{222}, \dots, y_{22r}}_{total = y_{22\cdot}}$	$\dots$	$\underbrace{y_{2b1}, y_{2b2}, \dots, y_{2br}}_{total = y_{2b\cdot}}$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
a	$\underbrace{y_{a11}, y_{a12}, \dots, y_{a1r}}_{total = y_{a1\cdot}}$	$\underbrace{y_{a21}, y_{a22}, \dots, y_{a2r}}_{total = y_{a2\cdot}}$	$\dots$	$\underbrace{y_{ab1}, y_{ab2}, \dots, y_{abr}}_{total = y_{ab\cdot}}$

Where, $y_{ij\cdot} = \sum_{k=1}^{r} y_{ijk}$ = the total of r observations of y for the treatment (ij).

Data Layout

Our table can then be rewritten as:

Factor B

Factor A	**1**	**2**	$\dots$	**b**	**Row Totals**	**Row Means**
1	$y_{11\cdot}$	$y_{12\cdot}$	$\dots$	$y_{1b\cdot}$	$y_{1\cdot\cdot} = \sum y_{1j\cdot}$	$\bar{y}_{1\cdot\cdot} = \frac{y_{1\cdot\cdot}}{rb}$
2	$y_{21\cdot}$	$y_{22\cdot}$	$\dots$	$y_{2b\cdot}$	$y_{2\cdot\cdot} = \sum y_{2j\cdot}$	$\bar{y}_{2\cdot\cdot} = \frac{y_{2\cdot\cdot}}{rb}$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$
a	$y_{a1\cdot}$	$y_{a2\cdot}$	$\dots$	$y_{ab\cdot}$	$y_{a\cdot\cdot} = \sum y_{aj\cdot}$	$\bar{y}_{a\cdot\cdot} = \frac{y_{a\cdot\cdot}}{rb}$
Column Totals	$y_{\cdot 1\cdot} = \sum y_{i1\cdot}$	$y_{\cdot 2\cdot} = \sum y_{i2\cdot}$	$\dots$	$y_{\cdot b\cdot} = \sum y_{ib\cdot}$	$y_{\dots}$	
Column Means	$\bar{y}_{\cdot 1\cdot} = \frac{y_{\cdot 1\cdot}}{ra}$	$\bar{y}_{\cdot 2\cdot} = \frac{y_{\cdot 2\cdot}}{ra}$	$\dots$	$\bar{y}_{\cdot b\cdot} = \frac{y_{\cdot b\cdot}}{ra}$		

Sum of Squares Formulas

- **Total:** $TSS = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r}(y_{ijk} - \bar{y}_{\dots})^2 =$
  $\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r} y_{ijk}^2 - \dfrac{(\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r} y_{ijk})^2}{rab}, \quad df = rab - 1 = n - 1$
- **Factor A:** $SSA = br\sum_{i=1}^{a}(\bar{y}_{i\cdot\cdot} - \bar{y}_{\dots})^2 = \sum_{i=1}^{a}\frac{y_{i\cdot\cdot}^2}{rb} - \dfrac{(\sum\sum\sum y_{ijk})^2}{rab}, \quad df = a - 1$
- **Factor B:** $SSB = ar\sum_{j=1}^{b}(\bar{y}_{\cdot j\cdot} - \bar{y}_{\dots})^2 = \sum_{j=1}^{b}\frac{y_{\cdot j\cdot}^2}{ra} - \frac{(\sum\sum\sum y_{ijk})^2}{rab}, \quad df = b - 1$
- **Interaction:** $SSAB = r\sum_{i=1}^{a}\sum_{j=1}^{b}(\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\dots})^2, \quad df = (a-1)(b-1)$
- **Treatment cell:**
  $SST_r = \sum_i\sum_j\sum_k(\bar{y}_{ij\cdot} - \bar{y}_{\dots})^2 = \sum_{i=1}^{a}\sum_{j=1}^{b}\frac{y_{ij\cdot}^2}{r} - \frac{(\sum\sum\sum y_{ijk})^2}{rab} \equiv SSA + SSB + SSAB$
- **Error:** $SSE = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r}(y_{ijk} - \bar{y}_{ij\cdot})^2, \quad df = ab(r-1)$
- **Check:** $TSS = SSA + SSB + SSAB + SSE \equiv SST_r + SSE$

Where,

$y_{ij\cdot} = \sum_{k=1}^{r} y_{ijk}$ = the total of r observations of y for the treatment (ij)
$y_{i\cdot\cdot} = \sum_{j=1}^{b} y_{ij\cdot} = \sum_{j=1}^{b}\sum_{k=1}^{r} y_{ijk}$, i.e. sum of all rb observations in row i
$\bar{y}_{i\cdot\cdot} = \frac{y_{i\cdot\cdot}}{rb}$ (estimator of $\mu_{i\cdot}$, i.e. mean of row i)
$y_{\cdot j\cdot} = \sum_{i=1}^{a} y_{ij\cdot} = \sum_{i=1}^{a}\sum_{k=1}^{r} y_{ijk}$, i.e. sum of all ra observations in column j
$\bar{y}_{\cdot j\cdot} = \frac{y_{\cdot j\cdot}}{ra}$ (estimator of $\mu_{\cdot j}$, i.e. mean of column j)
$y_{ij\cdot} = \sum_{k=1}^{r} y_{ijk} \implies \bar{y}_{ij\cdot} = \frac{y_{ij\cdot}}{r}$ (mean for cell ij)

And overall total $= y_{\dots} = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r} y_{ijk} \implies$ overall mean $= \bar{y}_{\dots} = \frac{y_{\dots}}{rab} = \frac{y_{\dots}}{n}$

ANOVA Table, F-Tests, and Decision Rules

Source	df	SS	MS	F
Factor A	$a-1$	SSA	$MSA = \frac{SSA}{a-1}$	$F_A = \frac{MSA}{MSE}$
Factor B	$b-1$	SSB	$MSB = \frac{SSB}{b-1}$	$F_B = \frac{MSB}{MSE}$
Interaction AB	$(a-1)(b-1)$	SSAB	$MSAB = \frac{SSAB}{(a-1)(b-1)}$	$F_{AB} = \frac{MSAB}{MSE}$
Error	$ab(r-1)$	SSE	$MSE = \frac{SSE}{ab(r-1)}$	
Total	$rab-1 = n-1$	TSS		

For a test at significance level α:

- **Interaction:** Reject H_0^{AB} if $F_{AB} > F_{(\alpha;(a-1)(b-1),ab(r-1))}$
- **Factor A:** Reject H_0^{A} if $F_A > F_{(\alpha;a-1,ab(r-1))}$
- **Factor B:** Reject H_0^{B} if $F_B > F_{(\alpha;b-1,ab(r-1))}$

Important: If H_0^{AB} is rejected (significant interaction), the main effects are usually not interpreted in isolation because the effect of A depends on the level of B.

Example: Supervisor Production Data

Example

Suppose that 2 supervisors were each observed on 3 randomly selected days for each of the 3 different shifts and the production outputs were recorded in a table below. Analyze this data

	Shift		
Supervisor	**Day** $= C_1$	**Swing** $= C_2$	**Night** $= C_3$
$1 = R_1$	571, 610, 625	480, 474, 540	470, 430, 450
$2 = R_2$	480, 516, 465	625, 600, 581	630, 680, 661

Table: Production outputs for 2 supervisors over 3 shifts with $r = 3$ replicates.

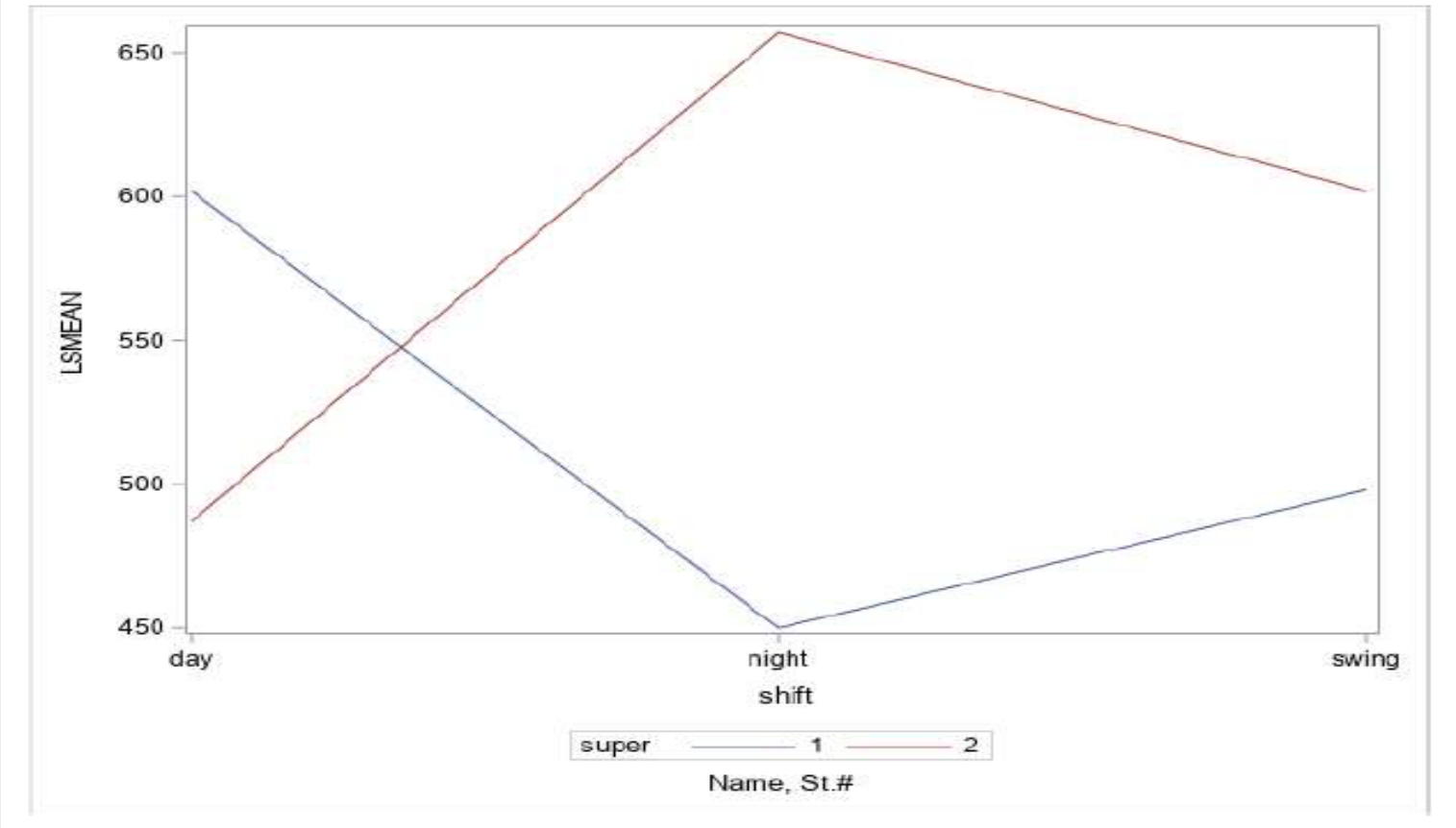
Here $a = 2, b = 3, r = 3$, and $n = ab = 6$. We also assume that the assumptions are valid: Normal distribution, with equal variance σ^2, and with constraints

$$\sum_{i=1}^{2} \alpha_i = \sum_{j=1}^{3} \beta_j = \sum_{i=1}^{2} (\alpha\beta)_{ij} \forall j = \sum_{j=1}^{3} (\alpha\beta)_{ij} \forall i = 0$$

Note: Row means $= R_i = \bar{y}_{i\cdot\cdot}$ and Column means $= C_j = \bar{y}_{\cdot j\cdot}$.

Example: Supervisor Production Data (Check for Interactions)

First, we need to check whether we have interactions between the 2 factors, by doing an interaction plot:



Example: Supervisor Production Data (Step-by-Step Calculations)

Next, we need to calculate all the sums of Squares and obtain the ANOVA table

$$TSS = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r} y_{ijk}^2 - \frac{\left(\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{r} y_{ijk}\right)^2}{rab}$$

$$= 5,541,030 - \frac{(9,888)^2}{18} = 109,222$$

$$SSA = \sum_{i=1}^{a} \frac{y_{i..}^2}{rb} - \frac{\left(\sum\sum\sum y_{ijk}\right)^2}{rab}$$

$$= \left(\frac{R_1^2}{9} + \frac{R_2^2}{9}\right) - \frac{(9,888)^2}{18} = 5,451,016 - \frac{(9,888)^2}{18} = 19,208$$

$$SSB = \sum_{j=1}^{b} \frac{y_{.j.}^2}{ra} - \frac{\left(\sum\sum\sum y_{ijk}\right)^2}{rab}$$

$$= \left(\frac{C_1^2}{6} + \frac{C_2^2}{6} + \frac{C_3^2}{6}\right) - \frac{(9,888)^2}{18} = 5,432,055 - \frac{(9,888)^2}{18} = 247$$

$$SSTr = \sum_{i=1}^{a}\sum_{j=1}^{b} \frac{y_{ij.}^2}{r} - \frac{\left(\sum\sum\sum y_{ijk}\right)^2}{rab}$$

$$= \left(\frac{(571+610+625)^2}{3} + \cdots + \frac{(630+680+661)^2}{3}\right) - \frac{(9,888)^2}{18} = 100,582$$

$$SS(AB) = SSTr - SSA - SSB = 81,127, \qquad SSE = TSS - SST_r - SSA - SSB = 81,127$$

Example: Supervisor Production Data (ANOVA Table)

Then, we calculate the mean squares and F-tests

$$MSA = \frac{SSA}{a-1} = 19,208, \quad MSB = \frac{SSB}{b-1} = 123.5, \quad MS(AB) = \frac{SS(AB)}{(a-1)(b-1)} = 40,563.5$$

$$MSE = \frac{SSE}{ab(r-1)} = \frac{8,640}{12} = 720$$

$$F_A = \frac{MSA}{MSE} = 26.677, \quad F_B = \frac{MSB}{MSE} = 0.17152, \quad F_{AB} = \frac{MS(AB)}{MSE} = 56.3382$$

The above results are then summarized in the ANOVA table:

Table: ANOVA Table for Two-Factor Factorial Design (Supervisor Production Data)

Source of Variation	**d.f.**	**SS**	**MS**	**F**
Treatments	5	100,582		
Factor A (Supervisor)	1	19,208	19,208	26.677
Factor B (Shift)	2	247	123.5	0.17152
AB (Interaction)	2	81,127	40,563.5	56.3382
Error	12	8,640	720	
Total	**17**	**109,222**		

R Output for Example

```
> data <- data.frame(Production=c(571, 610, 625, 480, 474, 540,
+ 470, 430, 450, 480, 516, 465, 625, 600, 581, 630, 680, 661),
+ Supervisor=rep(c("1","2"), each=9),
+ Shift=rep(rep(c("Day","Swing", "Night"), each=3),2))
> model <- aov(Production ~ Supervisor * Shift, data=data)
> summary(model)
Df Sum Sq Mean Sq F value   Pr(>F)
Supervisor        1  19208   19208  26.678 0.000235 ***
Shift             2    247     124   0.172 0.844406
Supervisor:Shift  2  81127   40564  56.338 7.95e-07 ***
Residuals        12   8640     720
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example: Supervisor Production Data (Test for Interactions)

Always test for interactions first:

$$H_0^{AB} : (\alpha\beta)_{ij} = 0 \quad \text{for all } i, j, \qquad \alpha = 0.05$$
$$H_a^{AB} : \text{ at least one } (\alpha\beta)_{ij} \neq 0.$$

Test statistic: $$F_{AB} = \frac{MS(AB)}{MSE} = 56.3382$$

R.R. Reject H_0 if $F_{AB} > F_{\alpha;(a-1)(b-1),\, ab(r-1)} = F_{0.05;(2,12)} = 3.89$

Since $F_{AB} = 56.3382 > 3.89$ we reject H_0 and conclude that at the 5% level of significance, i.e., there is evidence of an interaction between Supervisor and Shift.

Now we need to do all comparisons between treatment means μ_{ij} using Tukey's H.S.D. with $\binom{ab}{2} = \frac{ab(ab-1)}{2} = \binom{6}{2} = \frac{6(5)}{2} = 15$ comparisons.

Tukey's Procedure: Multiple Comparisons

If we reject any null hypothesis, we may want to compare specific means.

1. **For cell means:** $\left(\binom{ab}{2} = \frac{ab(ab-1)}{2} \text{ pairs}\right)$ $H_0 : \mu_{ij} = \mu_{\ell m}$ for all pairs $(i, j) \neq (\ell, m)$

$$\text{Critical value: } q_{(\alpha;ab,ab(r-1))}\sqrt{\frac{MSE}{r}}$$

$$\text{Difference in means : } \bar{y}_{ij\cdot} - \bar{y}_{\ell m\cdot}$$

2. **For marginal means of A:** $\left(\binom{a}{2} = \frac{a(a-1)}{2} \text{ pairs}\right)$ $H_0 : \mu_{i\cdot} = \mu_{\ell\cdot}$ for all $i \neq \ell$

$$\text{Critical value: } q_{(\alpha;a,ab(r-1))}\sqrt{\frac{MSE}{br}}$$

$$\text{Difference in means : } \bar{y}_{i\cdot\cdot} - \bar{y}_{\ell\cdot\cdot}$$

3. **For marginal means of B:** $\left(\binom{b}{2} = \frac{b(b-1)}{2} \text{ pairs}\right)$ $H_0 : \mu_{\cdot j} = \mu_{\cdot m}$ for all $j \neq m$

$$\text{Critical value: } q_{(\alpha;b,ab(r-1))}\sqrt{\frac{MSE}{ar}}$$

$$\text{Difference in means : } \bar{y}_{\cdot j\cdot} - \bar{y}_{\ell m\cdot}$$

Reject H_0 if $|\text{difference in means}| >$ critical value.

Tukey's Procedure: Simultaneous Confidence Intervals

The $1-\alpha$ simultaneous confidence intervals for all pairwise comparisons are given by:

1. **For cell means ($\mu_{ij} - \mu_{\ell m}$):**

$$(\bar{y}_{ij\cdot} - \bar{y}_{\ell m\cdot}) \pm q_{(\alpha;ab,ab(r-1))}\sqrt{\frac{MSE}{r}}$$

2. **For marginal means of A ($\mu_{i\cdot} - \mu_{\ell\cdot}$):**

$$(\bar{y}_{i\cdot\cdot} - \bar{y}_{\ell\cdot\cdot}) \pm q_{(\alpha;a,ab(r-1))}\sqrt{\frac{MSE}{br}}$$

3. **For marginal means of B ($\mu_{\cdot j} - \mu_{\cdot m}$):**

$$(\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot m\cdot}) \pm q_{(\alpha;b,ab(r-1))}\sqrt{\frac{MSE}{ar}}$$

Decision Rule: We are $1-\alpha$ confident that the true difference lies within the interval. If the interval does not contain zero, we reject H_0.

Example: Supervisor Production Data (Multiple Comparisons: Tukey)

Given the significant interaction ($F_{AB} = 56.34$, $p < 0.001$), we need to compare all treatment means μ_{ij}. There are $n = ab = 6$ treatment means, so $\binom{6}{2} = 15$ pairwise comparisons.

Tukey's Procedure:

- For comparing μ_{ij} vs $\mu_{\ell m}$, reject $H_0 : \mu_{ij} = \mu_{\ell m}$ if

$$|\bar{y}_{ij\cdot} - \bar{y}_{\ell m\cdot}| > q_\alpha(ab, ab(r-1))\sqrt{\frac{MSE}{r}}$$

- Here: $ab = 6$, $ab(r-1) = 12$, $\alpha = 0.05$, $MSE = 720$, $r = 3$
- $q_{0.05}(6, 12) = 4.75$ (from Studentized Range table)
- Critical value: $4.75 \times \sqrt{720/3} = 4.75 \times \sqrt{240} = 4.75 \times 15.4919 = 73.59$

First, calculate all treatment means $\bar{y}_{ij\cdot}$:

	Day (C_1)	**Swing** (C_2)	**Night** (C_3)
Supervisor 1 (R_1)	$\frac{571+610+625}{3} = 602$	$\frac{480+474+540}{3} = 498$	$\frac{470+430+450}{3} = 450$
Supervisor 2 (R_2)	$\frac{480+516+465}{3} = 487$	$\frac{625+600+581}{3} = 602$	$\frac{630+680+661}{3} = 657$

Denote:

$$\bar{y}_{11} = 602 \quad \bar{y}_{12} = 498 \quad \bar{y}_{13} = 450$$
$$\bar{y}_{21} = 487 \quad \bar{y}_{22} = 602 \quad \bar{y}_{23} = 657$$

Example: Supervisor Production Data (Comparison 1: μ_{11} vs μ_{12})

Hypothesis: $H_0 : \mu_{11} = \mu_{12}$ vs $H_1 : \mu_{11} \neq \mu_{12}$
Test Statistic:

$$|\bar{y}_{11} - \bar{y}_{12}| = |602.00 - 498.00| = 104.00$$

Decision: Since $104.00 > 73.59$, reject H_0.
Simultaneous 95% Confidence Interval:

$$(\bar{y}_{11} - \bar{y}_{12}) \pm q_{0.05}(6, 12)\sqrt{\frac{MSE}{r}}$$

$$104.00 \pm 73.59$$

$$[30.41, 177.59]$$

Since 0 is not in the interval, $\mu_{11} \neq \mu_{12}$.

Example: Supervisor Production Data (Comparison 2: μ_{11} vs μ_{13})

Hypothesis: $H_0 : \mu_{11} = \mu_{13}$ vs $H_1 : \mu_{11} \neq \mu_{13}$
Test Statistic:

$$|\bar{y}_{11} - \bar{y}_{13}| = |602.00 - 450.00| = 152.00$$

Decision: Since $152.00 > 73.59$, reject H_0.
Simultaneous 95% CI:

$$152.00 \pm 73.59 = [78.41, 225.59]$$

Example: Supervisor Production Data (Summary of Tukey's Test Results)

Comparison	Treatment	Diff	Sig?	95% CI Lower	95% CI Upper
μ_{11} vs μ_{12}	Sup. 1, Day vs Sup. 1, Swing	104.00	Yes	30.41	177.59
μ_{11} vs μ_{13}	Sup. 1, Day vs Sup. 1, Night	152.00	Yes	78.41	225.59
μ_{11} vs μ_{21}	Sup. 1, Day vs Sup. 2, Day	115.00	Yes	41.41	188.59
μ_{11} vs μ_{22}	Sup. 1, Day vs Sup. 2, Swing	0.00	No	-73.59	73.59
μ_{11} vs μ_{23}	Sup. 1, Day vs Sup. 2, Night	55.00	No	-18.59	128.59
μ_{12} vs μ_{13}	Sup. 1, Swing vs Sup. 1, Night	48.00	No	-25.59	121.59
μ_{12} vs μ_{21}	Sup. 1, Swing vs Sup. 2, Day	11.00	No	-62.59	84.59
μ_{12} vs μ_{22}	Sup. 1, Swing vs Sup. 2, Swing	104.00	Yes	30.41	177.59
μ_{12} vs μ_{23}	Sup. 1, Swing vs Sup. 2, Night	159.00	Yes	85.41	232.59
μ_{13} vs μ_{21}	Sup. 1, Night vs Sup. 2, Day	37.00	No	-36.59	110.59
μ_{13} vs μ_{22}	Sup. 1, Night vs Sup. 2, Swing	152.00	Yes	78.41	225.59
μ_{13} vs μ_{23}	Sup. 1, Night vs Sup. 2, Night	207.00	Yes	133.41	280.59
μ_{21} vs μ_{22}	Sup. 2, Day vs Sup. 2, Swing	115.00	Yes	41.41	188.59
μ_{21} vs μ_{23}	Sup. 2, Day vs Sup. 2, Night	170.00	Yes	96.41	243.59
μ_{22} vs μ_{23}	Sup. 2, Swing vs Sup. 2, Night	55.00	No	-18.59	128.59

Another Real-Life Example: 2×2 Factorial

Example

A researcher studies the effect of **Exercise Type** (A: Running, Cycling) and **Diet** (B: Low-carb, Low-fat) on weight loss (kg) over 12 weeks. $r = 4$ subjects per treatment.

	Low-carb ($j = 1$)				Low-fat ($j = 2$)			
Running ($i = 1$)	4.2,	4.5,	4.0,	4.3	3.0,	3.2,	2.8,	3.1
Cycling ($i = 2$)	3.8,	3.9,	3.7,	4.0	2.5,	2.7,	2.3,	2.6

Step-by-Step Calculations (1)

1. Calculate cell means:

$$\bar{y}_{11\cdot} = (4.2 + 4.5 + 4.0 + 4.3)/4 = 4.25$$
$$\bar{y}_{12\cdot} = (3.0 + 3.2 + 2.8 + 3.1)/4 = 3.025$$
$$\bar{y}_{21\cdot} = (3.8 + 3.9 + 3.7 + 4.0)/4 = 3.85$$
$$\bar{y}_{22\cdot} = (2.5 + 2.7 + 2.3 + 2.6)/4 = 2.525$$

2. Calculate marginal means:

$$\bar{y}_{1\cdot\cdot} = (4.25 + 3.025)/2 = 3.6375$$
$$\bar{y}_{2\cdot\cdot} = (3.85 + 2.525)/2 = 3.1875$$
$$\bar{y}_{\cdot 1\cdot} = (4.25 + 3.85)/2 = 4.05$$
$$\bar{y}_{\cdot 2\cdot} = (3.025 + 2.525)/2 = 2.775$$

3. Overall mean: $\bar{y}_{\cdots} = (3.6375 + 3.1875)/2 \overset{\text{or}}{=} (4.05 + 2.775)/2 = 3.4125$

Step-by-Step Calculations (2)

4. Calculate sums of squares:

$$
\begin{aligned}
SSA &= 4 \times 2 \times [(3.6375 - 3.4125)^2 + (3.1875 - 3.4125)^2] \\
&= 8 \times (0.0506 + 0.0506) = 0.8096 \approx 0.810 \\
SSB &= 4 \times 2 \times [(4.05 - 3.4125)^2 + (2.775 - 3.4125)^2] \\
&= 8 \times (0.4064062 + 0.4064063) = 6.5025 \approx 6.503 \\
SSAB &= 4 \times [(4.25 - 3.6375 - 4.05 + 3.4125)^2 + (3.025 - 3.6375 - 2.775 + 3.4125)^2 \\
&+ (3.85 - 3.1875 - 4.05 + 3.4125)^2 + (2.525 - 3.1875 - 2.775 + 3.4125)^2] \\
&= 4 \times [0.000625 + 0.000625 + 0.000625 + 0.000625] = 0.010 \\
TSS &= \sum (y_{ijk} - 3.4125)^2 = (4.2 - 3.4125)^2 + (4.5 - 3.4125)^2 \\
&+ (4.0 - 3.4125)^2 + (4.3 - 3.4125)^2 + (3 - 3.4125)^2 + (3.2 - 3.4125)^2 \\
&+ (2.8 - 3.4125)^2 + (3.1 - 3.4125)^2 + (3.8 - 3.4125)^2 + (3.9 - 3.4125)^2 \\
&+ (3.7 - 3.4125)^2 + (4 - 3.4125)^2 + (2.5 - 3.4125)^2 + (2.7 - 3.4125)^2 \\
&+ (2.3 - 3.4125)^2 + (2.6 - 3.4125)^2 = 2.9357 + 0.6881 + 0.8156 + 3.2381 \\
&= 7.6775 \approx 7.678 \\
SSE &= TSS - SSA - SSB - SSAB \\
&= 7.678 - 0.810 - 6.503 - 0.010 = 0.355
\end{aligned}
$$

Step-by-Step Calculations (3)

5. Complete ANOVA table:

Source	df	SS	MS	F
Exercise Type (A)	1	0.810	0.810	27.36
Diet (B)	1	6.503	6.503	219.59
Interaction (AB)	1	0.010	0.010	0.338
Error	12	0.355	0.0296	
Total	15	7.678		

6. Critical values ($\alpha = 0.05$): $F_{0.05;1,12} = 4.75$
7. Decisions:
   - Interaction: $F = 27.36 > 4.75 \rightarrow$ Reject H_0^{AB} (there is significant interaction)
   - Exercise Type: $F = 219.59 > 4.75 \rightarrow$ Reject H_0^{A} (significant)
   - Diet: $F = 0.338 < 4.75 \rightarrow$ Fail to reject H_0^{B} (not significant)

Summary

- Two-factor factorial designs allow simultaneous study of two factors and their interaction.
- The full model includes main effects and interaction term.
- Always test interaction first. If significant, interpret effects cautiously.
- ANOVA decomposes total variation into components for A, B, AB, and error.
- Use Tukey's procedure for multiple comparisons if needed.